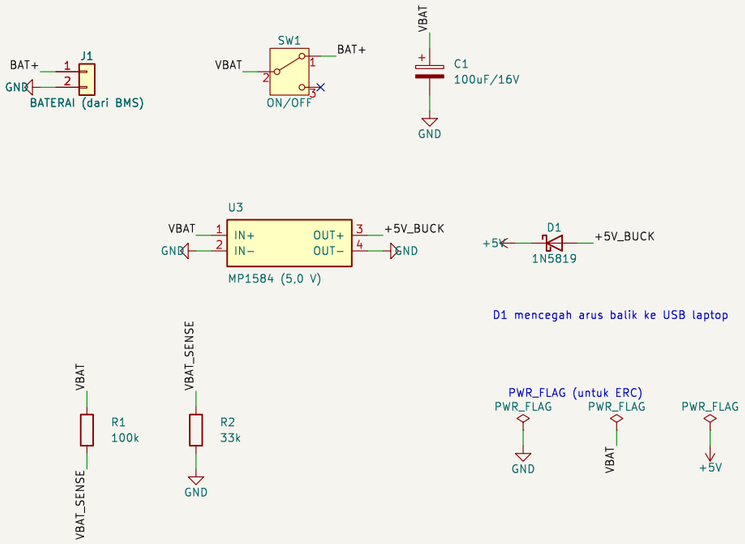
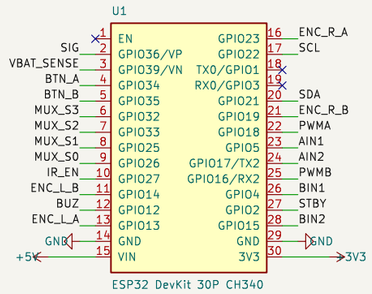
1. DAYA: baterai, saklar, buck 5 V, pembaca baterai

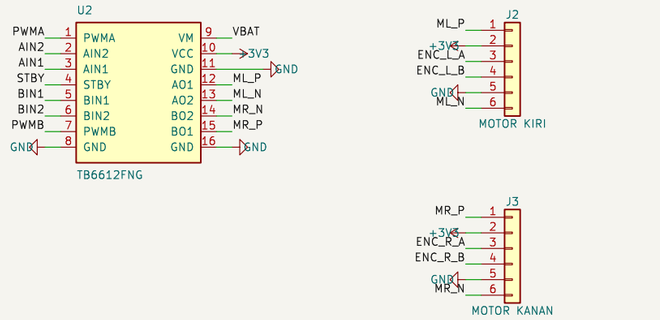


2. ESP32 DevKit 30 pin



GPIO12 (BUZ) harus LOW saat boot: ada R4 10k di basis Q1

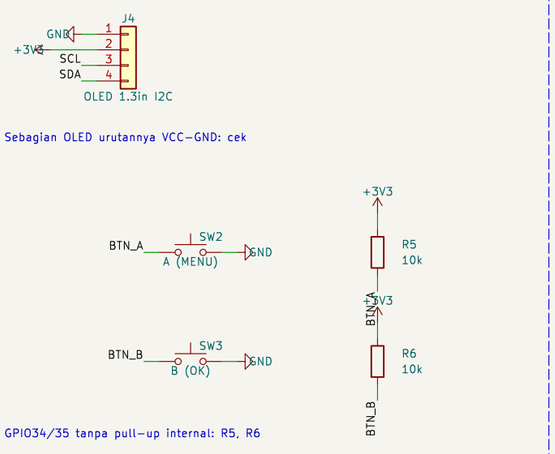
3. DRIVER MOTOR + KONEKTOR MOTOR



Urutan kabel N20 beda tiap penjual: cocokkan

Encoder: pakai pull-up internal ESP32 (INPUT_PULLUP)

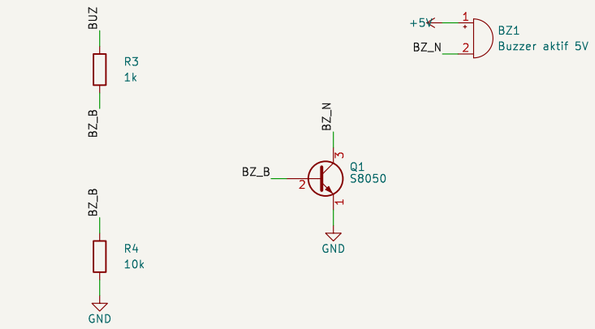
4. OLED + TOMBOL



Sebagian OLED urutannya VCC–GND: cek

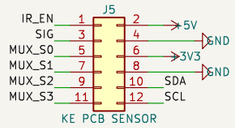
GPIO34/35 tanpa pull-up internal: R5, R6

5. BUZZER



Buzzer magnetik: tambah 1N4148 paralel

6. KE PCB SENSOR (kabel pita IDC 12 pin)



Urutan pin sama persis dengan J1 PCB sensor

Mux: SIG GPIO36, S0–S3 GPIO32/33/2/15
LED sensor: MCP23017 di bus I2C (0x20)

7. LUBANG BAUT M3



H1–H2 rangka, H3–H4 bahu + dudukan PCB sensor

v2 (13/10/2026): disederhanakan, semua komponen kaki tembus
Pin modul = susunan umum, CEK dengan modul yang dibeli
Acuan: docs/skematik-v1.md
P01 Kel.5 PjBL DTEO FV ITS

Sheet: /		
File: ltm-main.kicad_sch		
Title: LTM – PCB Utama		
Size: A3	Date: 2026-10-13	**Rev: v2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1